

International Institute of Information Technology, Bhubaneswar

Department of Computer Science and Engineering

Object Oriented Programming Laboratory

Lab 6 – Test on C++ Pointers (Set B)

Date: 01.09.2026**Semester:** B.Tech 3rd Semester**Group:** 2**Instructions**

- Write all programs in C++.
- Use **pointers** wherever required in each question.
- Use pointer dereferencing and pointer arithmetic appropriately.
- Do not use array indexing where explicitly restricted.
- Properly release dynamically allocated memory using `delete` or `delete[]`.
- Display the output in a clear and readable format.

1. Mobile Battery Update

A mobile phone stores its battery percentage in a variable.

Create a pointer to the battery percentage. Using the pointer:

1. Display the current battery percentage.
2. Increase it after charging.
3. Display the updated percentage.

Condition: Modify the battery percentage only through the pointer.

2. Water Tank Level

A smart water tank stores its current water level.

Create a pointer pointing to the water level. Using the pointer:

1. Display the current level.
2. Add an amount of water.
3. Remove an amount of water.
4. Display the final water level.

3. Sports Equipment Rack

A sports centre stores the identification numbers of 6 pieces of equipment in an array.

Using a pointer:

1. Display all equipment IDs.
2. Display the address of each element.

Condition: Move through the array using pointer arithmetic.

4. Train Seat Correction

A railway system stores 8 seat numbers in an array.

A seat number at a position entered by the user was recorded incorrectly.

Use pointer arithmetic to update that seat number.

Display the list before and after correction.

Condition: Do not use `arr[position]`.

5. Online Order Status

An online shopping system stores an order status code:

- 1 = Processing
- 2 = Shipped
- 3 = Delivered

Write a function:

```
void updateStatus(int *status);
```

The function should update:

- Processing → Shipped
- Shipped → Delivered

Display the status before and after calling the function.

6. Podcast Duration Analyzer

A podcast application stores the duration of 6 episodes.

Write a function that receives:

- A pointer to the first duration.
- The number of episodes.

Using pointer traversal, find and display the longest episode duration.

Condition: Do not use array indexing inside the function.

7. Text Analyzer

A note-taking application stores a sentence in a character array.

Using a character pointer, count:

- Number of digits.
- Number of alphabetic characters.
- Number of spaces.

Traverse the string until `'\0'`.

8. Classroom Marks Update

A teacher stores the marks of n students in an array.

Write a function that receives a pointer to the marks and the number of students. The function should add 5 marks to every student.

Display the marks before and after modification.

Condition: Modify the original array using pointers.

9. Restaurant Table Manager

A restaurant does not know in advance how many table numbers will be entered.

Write a program that:

1. Dynamically allocates memory for n table numbers.
2. Accepts all table numbers.
3. Finds the smallest table number using pointer traversal.
4. Releases the dynamically allocated memory.

10. Contact Number Search

A company stores a variable number of contact numbers.

Write a program that:

1. Dynamically allocates memory for n contact numbers.
2. Accepts the contact numbers.
3. Searches for a specified contact number using a pointer.
4. Displays whether it is found and its position.
5. Properly deallocates the allocated memory.

Condition: Do not use array indexing while searching.